Syllabus:

Course Contents	
Unit-I	**Water Treatment** Introduction, types of hardness; Industrial treatment of water , 1) lime soda process 2) process 3) Ion-exchange process .Numerical based on lime-soda and Zeolite process. Boiler troubles: 1) sludge and scale formation 2) Caustic Embrittlement 3) Boiler Corrosion 4) Priming and Foaming. Conditioning of

Scanned with OKEN Scanner

	water :1) Carbonate 2) Calgon 3) Phosphate .Domestic Treatment of drinking water **[6 Hrs]**
Unit-II	**Nanomaterials** General introduction to nanoscience. Methods of synthesis of nanomaterials: 'Top-Down' and 'Bottom-Up', Carbon nanotubes: single-walled and multi-walled carbon nanotubes, their structures, properties and applications. use of nanomaterials in electronics, sensors, catalysis, environment **[5 Hrs]**
Unit-III	**Periodic Properties** Effective nuclear charge, electronic configurations, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electro negativity, polarizability, oxidation states, hard soft acids and bases **[6 Hrs]**
Unit-IV	**A) Atomic and molecular structure:** Molecular orbitals of diatomic molecules and plots of the multi center orbitals. Equations for atomic and molecular orbital. Energy level diagrams of diatomic(hydrogen and oxygen) Formation of wave function (hydrogen) , Benzene structure **B) Energy:-** Introduction, Analysis of coal-Proximate analysis, liquid fuel-Refining of Crude oil and Gaseous fuel, Combustion Calculation **[7 Hrs]**
Unit-V	**A) Advanced Polymeric Materials :-** liquid crystal polymers (thermotropic and lyotropic), phases of thermotropic polymers: nematic, smectic, cholesteric; advantages, disadvantages and applications **B) Spectroscopic Techniques and Applications:-** UV spectroscopy :- Introduction, Beers Lamberts law, Types of transition, concept of auxochrome and chromophore NMR spectroscopy :- Instrumentation , Number of signals (Equivalence and Non - equivalent Protons) , Solvents used in NMR, Chemical shift, Application of NMR spectroscopy **[6 Hrs]**
Text Books	
1	Textbook of Engineering Chemistry, Dr.S.S.Dara, Dr.S.S.Umre, S.Chand and Company Ltd., Twelfth/2011
2	Nanomaterials, Nanotechnology and Design, Michael F. Ashby, Paulo J. Ferreira, DanielL. Schodek, Elsevier, First/2013
Reference Books	
1	Engineering Chemistry, P.C.Jain And MonikaJain, Dhanpatrai Publishing Company Ltd., 15thEd/2009
2	Principles of Physical Chemistry, B.R.Puri , L.R.Sharma and Madan S. Pathania,Vishal Publishing Company, First/2002
3	Chemistry, John E McMurry and Robert C Fay, Pearson, First/2008,
4	Nanotechnology Agent le Introduction to the Next big Idea,Mark Ratner, Daniel Ratner, Pearson, First/2017
Useful links	
1	https://nptel.ac.in/courses/105106119/ (time : from 19.55 to 38.05)
2	https://nptel.ac.in/courses/103103026/pdf/mod4.pdf

Scanned with OKEN Scanner